

Basic Statistical Descriptions of Data

1 Motivation

To better understand data, we must analyze its **central tendency**, **variation**, and **spread**. Data dispersion characteristics—such as the median, max, min, quantiles, and outliers—provide a comprehensive view of the dataset's distribution.

- **Numerical Dimensions:** Correspond to sorted intervals.
- **Granularity:** Dispersion can be analyzed at multiple levels of precision.
- **Techniques:** Boxplots, quantile analysis, and dispersion analysis on computed measures.

2 Measuring Central Tendency

Central tendency identifies the "center" or typical value of a data distribution.

2.1 Mean (Arithmetic Average)

The most common measure of central tendency.

Mean Formulas

For a **Sample** of size n :

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

Where: x_i is each individual value, and n is the total number of values in the sample.

For a **Population** of size N :

$$\mu = \frac{\sum_{i=1}^N x_i}{N}$$

Where: x_i is each value in the population, and N is the total population size.

- **Weighted Arithmetic Mean:** $\bar{x} = \frac{\sum w_i x_i}{\sum w_i}$
Where: w_i is the weight of each value x_i .
- **Trimmed Mean:** Calculated by "chopping off" extreme values (outliers) to avoid skewing.

2.2 Median

The middle value in a sorted data set.

- **Odd n :** The middle value.
- **Even n :** The average of the two middle values.

Estimated Median (Grouped Data)

For grouped data, the median can be estimated by interpolation:

$$\text{Median} = L_1 + \left(\frac{\frac{n}{2} - (\sum f)_l}{f_{\text{median}}} \right) c$$

Where L_1 is the lower boundary of the median class, $(\sum f)_l$ is the sum of frequencies of all classes lower than the median class, f_{median} is the frequency of the median class, and c is the class interval width.

2.3 Mode

The value that occurs most frequently.

- **Unimodal:** One mode.
- **Bimodal / Trimodal:** Two or three modes.

Empirical Formula (Mode)

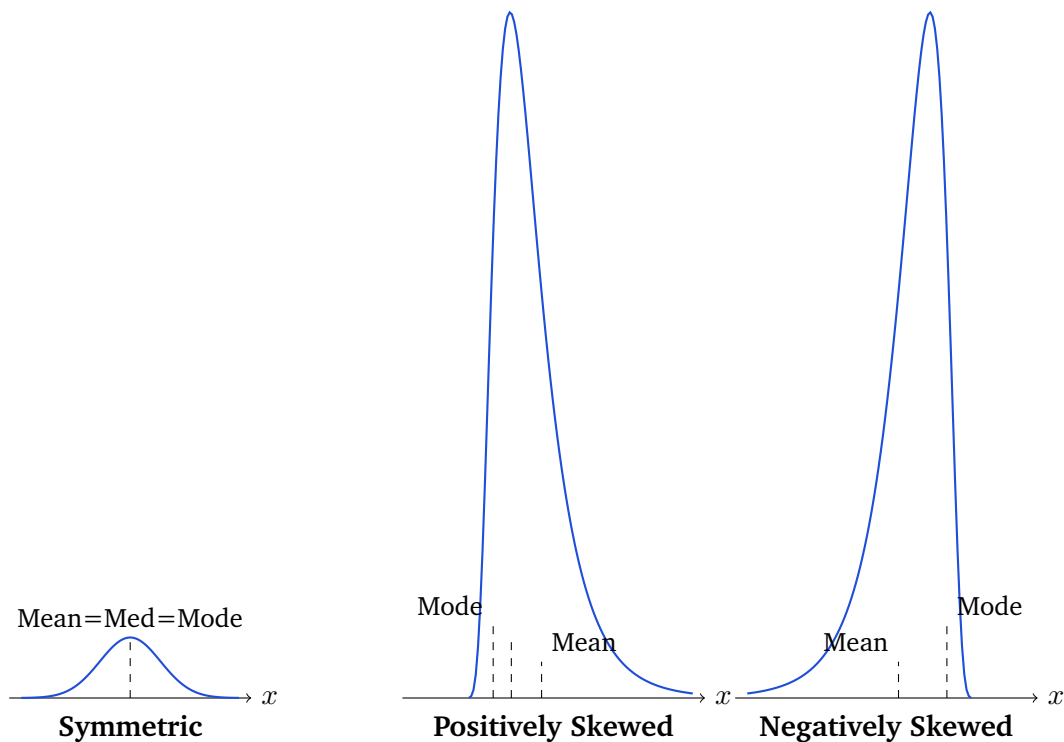
For moderately skewed distributions:

$$\text{Mode} \approx 3 \times \text{Median} - 2 \times \text{Mean}$$

Alternatively: $\text{Mean} - \text{Mode} \approx 3 \times (\text{Mean} - \text{Median})$

3 Symmetry and Skewness

The relationship between Mean, Median, and Mode describes the "shape" of the distribution.



Type	Relationship	Example Configuration
Symmetric	Mean = Med = Mode	{40, 50, 60, 70, 80} → Mean=60, Med=60
Positive Skew	Mean > Med > Mode	{200, 300, 350, 400, 450, 1200} (High income outliers)
Negative Skew	Mean < Med < Mode	{95, 90, 88, 85, 60} (A few low test scores)

4 Measuring Data Dispersion

4.1 Range and Quantiles

- **Range:** Difference between the Max and Min values.
- **Quantiles:** Points taken at regular intervals that divide the sorted data into equal-size sets.

Common Quantiles

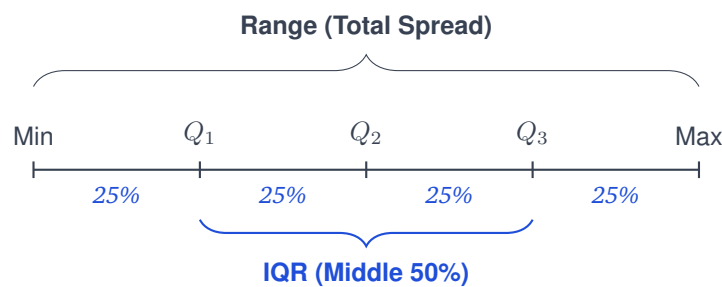
- **Quartiles (4-quantiles):** Q_1 (25th), Q_2 (Median, 50th), Q_3 (75th).
- **Percentiles (100-quantiles):** Divide data into 100 parts.

4.2 Interquartile Range (IQR)

The range covered by the middle 50% of the data:

$$\text{IQR} = Q_3 - Q_1$$

Where: Q_3 is the third quartile (75th percentile) and Q_1 is the first quartile (25th percentile).

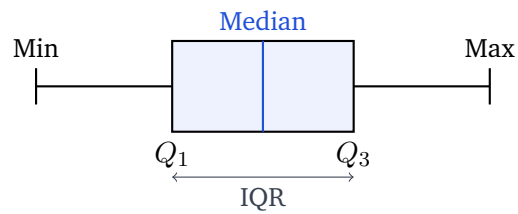


Visual Representation: Quantiles divide the sorted data distribution into consecutive sets of equal size.

5 Boxplots and Five-Number Summary

A boxplot visualizes the **Five-Number Summary**:

$$\{\text{Minimum}, Q_1, \text{Median}, Q_3, \text{Maximum}\}$$



- **Whiskers:** Extend to the Min and Max.
- **Outliers:** Points beyond the whiskers (typically $> 1.5 \times \text{IQR}$ from the box).